

HDOC DAY-35

Name-Priyadarshna

Sapid-590032928

Batch-15

QUESTION:

Write a menu-driven C program to read a square matrix and perform the following operations using separate functions:

1. Check whether all main diagonal elements are distinct.
2. Find the sum of main diagonal elements.
3. Display the elements using diagonal traversal.
4. Exit.

Aim:

To write a menu-driven C program to perform operations on a square matrix using separate functions:

1. Check whether the main diagonal elements are distinct.
2. Find the sum of main diagonal elements.
3. Display the matrix elements using diagonal traversal.
4. Exit the program.

Algorithm

1. Start.
2. Read the order n of the square matrix.
3. Read all matrix elements.
4. Display the menu.
5. Read the user's choice.
6. If choice is **1**, check whether main diagonal elements are distinct.
7. If choice is **2**, calculate and display the sum of main diagonal elements.
8. If choice is **3**, display elements using diagonal traversal.
9. If choice is **4**, exit the program.

10. Otherwise, display **Invalid choice**.
11. Repeat the menu until Exit is selected.
12. Stop.

Test Case	Matrix	Choice	Expected Output
1	1 2 3 / 4 5 6 / 7 8 9	1	Main diagonal elements are distinct
2	1 2 3 / 4 5 6 / 7 8 9	2	Sum = 15
3	1 2 3 / 4 5 6 / 7 8 9	3	1 2 4 7 5 3 6 8 9
4	Any matrix	4	Exit

>C program:

```

1 #include <stdio.h>
2
3 void distinct(int n, int a[n][n]) {
4     int i, j, flag = 1;
5     for(i = 0; i < n; i++)
6         for(j = i + 1; j < n; j++)
7             if(a[i][i] == a[j][j])
8                 flag = 0;
9
10    if(flag)
11        printf("Main diagonal elements are distinct.\n");
12    else
13        printf("Main diagonal elements are not distinct.\n");
14}
15
16 void sumDiagonal(int n, int a[n][n]) {
17     int i, sum = 0;
18     for(i = 0; i < n; i++)
19         sum += a[i][i];
20
21     printf("Sum of main diagonal = %d\n", sum);
22}
23
24 void diagonalTraversal(int n, int a[n][n]) {
25     int s, i, j;
26
27     for(s = 0; s <= 2*n-2; s++) {
28         if(s % 2 == 0) {
29             i = (s < n) ? s : n-1;
30             j = s - i;
31             while(i >= 0 && j < n)
32                 printf("%d ", a[i--][j++]);
33         }
34         else {
35             j = (s < n) ? s : n-1;
36             i = s - j;
37             while(j >= 0 && i < n)
38                 printf("%d ", a[i++][j--]);
39         }
40     }
41     printf("\n");
42}
43
44 int main() {
45     int n, i, j, choice;
46
47     printf("Enter order of matrix: ");

```

```
44 int main() {
45     int n, i, j, choice;
46
47     printf("Enter order of matrix: ");
48     scanf("%d", &n);
49
50     int a[n][n];
51
52     printf("Enter matrix elements:\n");
53     for(i = 0; i < n; i++)
54         for(j = 0; j < n; j++)
55             scanf("%d", &a[i][j]);
56
57     do {
58         printf("\n1. Check distinct diagonal");
59         printf("\n2. Sum of main diagonal");
60         printf("\n3. Diagonal traversal");
61         printf("\n4. Exit");
62         printf("\nEnter choice: ");
63         scanf("%d", &choice);
64
65         switch(choice) {
66             case 1: distinct(n, a); break;
67             case 2: sumDiagonal(n, a); break;
68             case 3: diagonalTraversal(n, a); break;
69             case 4: printf("Exit\n"); break;
70             default: printf("Invalid choice\n");
71         }
72     } while(choice != 4);
73
74     return 0;
75 }
76
```

>OUTPUT:

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$ gcc day35.c -o day35
>
[priyadarshna@kali]~$ ./day35
Enter order of matrix: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9

1. Check distinct diagonal
2. Sum of main diagonal
3. Diagonal traversal
4. Exit
Enter choice: 1
Main diagonal elements are distinct.

1. Check distinct diagonal
2. Sum of main diagonal
3. Diagonal traversal
4. Exit
Enter choice: 2
Sum of main diagonal = 15

1. Check distinct diagonal
2. Sum of main diagonal
3. Diagonal traversal
4. Exit
Enter choice: 3
1 2 4 7 5 3 8 6 9

1. Check distinct diagonal
2. Sum of main diagonal
3. Diagonal traversal
4. Exit
Enter choice: 4
Exit
[priyadarshna@kali]~$
```